Nalin Agarwal

+44 7770025803 | nagarwal@ed.ac.uk | LinkedIn | GitHub | nalinagarwal.com | Edinburgh, Scotland, UK

Background

AI and robotics researcher specialising in robot learning, dexterous manipulation, tactile sensing and sim-to-real reinforcement learning. Interested in embodied cognition and how robots develop perception, memory and self-models, with the long-term goal of building embodied systems that could develop subjective experience and a sense of self.

Education

The University of Edinburgh — Edinburgh, Scotland, UK

M.Sc. in Artificial Intelligence; ***Grade: Merit (70.6%)*** — *Sept 2025 – Sept 2026*

Specialising in Robotics and RL, including the master's dissertation;

Relevant coursework: Advanced Robotics, Robot and Reinforcement Learning, Probabilistic Modeling and Reasoning, Advanced topics in Machine Learning, Computational Neuroscience, Computational Cognitive Neuroscience, Introduction to Quantum Computing, Computer Vision

Delhi Technological University — New Delhi, India

B.Tech. in Engineering Physics; ***GPA: 9.18/10*** — *Dec 2020 – May 2024*

Major in Electronics and Communications Engineering, Minor in Computer Science Engineering;

Relevant coursework: Fundamental Physics, Mathematics, MATLAB Applied Physics, Quantum Physics, Cosmology and Astrophysics, Electronics, Semiconductors, Optical Fibres, Microwave, Atmospheric Sciences, Data Structures and Algorithms, Database Systems, Operating Systems, Computer Networks, Computer Graphics

Skills

- **Robotics:** ROS/ROS2, Isaac Lab, MuJoCo, Pinocchio.
- **Hardware:** Arduino, microcontroller interfacing, tactile sensing.
- **Machine Learning:** PyTorch, Reinforcement Learning (PPO, SAC), Computer Vision, PennyLane, Qiskit.
- **Programming and Tools:** Python, C/C++, MATLAB, SQL, Git, Linux, LaTeX, HTML, CSS, JavaScript, Flask, Jinja, Godot, Unity, Adobe Creative Suite, Silvaco TCAD.

Research Experience

Statistical Machine Learning and Motor Control (SLMC) Group, Institute of Perception, Action and Behaviour (IPAB) — Edinburgh, Scotland, UK

Research Assistant, University of Edinburgh — *Oct 2026 – Present, Full-time*

- Developing MuJoCo simulation infrastructure for robotic box folding, focusing on real-time performance, object-state resets and trajectory execution.
- Building workflows for controller-gain tuning, parallel parameter sweeps with mjbatch, and integration with reinforcement-learning and data-collection pipelines.
- Developing ROS2 interfaces for NEXTAGE trajectory editing and folding execution, with streaming teleoperation and simultaneous publishing from multiple robot cameras.
- Working on MuJoCo model integration and controller stabilisation for the AIREC robot.

Student Researcher, University of Edinburgh — *Jan 2026 – Sept 2026, MSc Dissertation*

- Worked on the MSc Dissertation titled - **"Development of a Low-cost Tactile System for Dexterous Manipulation and Investigating the Sim-To-Real Gap in Tactile Reinforcement Learning"** under the supervision of Prof. Sethu Vijayakumar and Elle Miller.
- Developed a low-cost tactile sensing system for the Shadow Hand Lite, integrating 12 FSR sensors with four BioTac fingertip sensors and implementing a per-channel hysteresis calibration along with PadTac USD assets in Isaac Lab and a ROS deployment pipeline for blind Baoding manipulation using PPO policies at 60 Hz.
- Developed slew-rate domain randomisation to improve sim-to-real transfer by accounting for hardware motion limits and encouraging smoother policy commands. Achieved zero-shot hardware transfer completing **112.5 consecutive full rotations** without dropping a ball; the final policy averaged 34.75 full rotations across 10 benchmark trials.

Centre for Development of Advanced Computing (C-DAC) — Pune, India

Research Assistant — *Nov 2024 – March 2025, Full-time*

- Worked in the field of Quantum Computing with specific focus on Quantum Artificial Intelligence and QML algorithms.
- Briefly studied various Quantum Machine Learning Algorithms for optimisation techniques like Quantum Approximate Optimisation Algorithm (QAOA) and Variational Quantum Eigensolver (VQE), and their different range of applications in recent R&D developments.

- Learned the programming techniques for Qiskit and used Hybrid Quantum Convolutional Neural Networks for image-based recognition and detection of Wildfires in forest areas.

Indian Institute of Technology, Delhi (IIT-D) New Delhi, India
Research Fellowship *Feb 2024 – Oct 2024, Part-Time*

- Worked at the Centre for Atmospheric Sciences (CAS) department at IIT-Delhi under the supervision of Dr. Samiran Mandal on the project "Computational Analysis of oceanic events across the Seychelles-Chagos Thermocline Ridge region in the Southern Indian Ocean"
- Data Analysis using Python programming language to find heatwave hotspots
- Physical analysis of various metrics and correlation with IOD/ENSO events, sea surface height, chlorophyll anomaly across the specific region using mathematical tools, and computing the data to draw comparisons and conclusions.

Space Applications Centre, ISRO Ahmedabad, India
Research Assistant *May 2022 – Jul 2022 ; Jun 2023 - Aug 2023, Full-time*

- Worked on "Analysis of Marine Heatwaves Using Satellite Data" under the supervision of Dr. Rashmi Sharma and Dr. Nitant Dube.
- Data Analysis of satellite data using Python programming language
- Development of a tool to detect real-time Marine Heatwaves over the globe based on satellite data over the past 40 years. Analysed various Marine Heatwaves metrics using python plots and studied correlation of cyclones and marine heatwaves in the Indian subcontinent ocean in detail.

Delhi Technological University New Delhi, India
Undergraduate Researcher *Aug 2023 – May 2024, Part-Time*

- Worked on the undergraduate thesis project in the field of Semiconductor Electronics incorporating device/circuit design and simulations using Silvaco TCAD software and ATLAS language.
- Studied different possible structural and material changes to the dual gate MOSFET to incorporate charge trapping capabilities.
- Published an article about the research findings in an IEEE conference as well.

Publications

IEEE International Conference on Robotics and Automation (ICRA) 2027 *Under review*
Title: No Eyes, No Problem: What Does Touch Contribute to Blind Baoding Ball Manipulation?
Authors: Ayush Deshmukh*, **Nalin Agarwal***, Elle Miller, Sethu Vijayakumar
**Equal contribution*
Project Page : **No Eyes, No Problem** — **arXiv**

Conference on Robot Learning (CoRL) 2026
Title: No Eyes, No Problem: Zero-Shot Blind Baoding Ball Manipulation with Binary Touch
Authors: Ayush Deshmukh*, **Nalin Agarwal***, Elle Miller, Sethu Vijayakumar
**Equal contribution*

Workshops:
3rd Workshop on Dexterous Manipulation: Full-Stack Robotics on Learning and Design *Under review*
Open Problems in Contact-Rich Loco-Manipulation *Under review*

ICITEICS 2024, Vemana Institute of Technology - Koramangala, Bengaluru *July 2024*
Title : A Novel Dielectric modulated Double Gate Charge-Trapping MOSFET: Ferroelectric Gate Stack and Gaussian Doped Substrate Integration for Enhanced Performance
Authors: Akkshita Swain, **Nalin Agarwal**, Tanisha Singhal, Rishu Chaujar
Certificate : **Link**
DOI: 10.1109/ICITEICS61368.2024.10625402
Published By: IEEE

TROPMET-2022, Indian Institute of Science Education and Research, Bhopal *Dec 2022*
Title : "On Rapid Increase In The MHW Intensity During Last Decade (2010-Present) Using Satellite Observations: Implications on Cyclone Activity in the Arabian Sea?
Authors: **Nalin Agarwal**, Nitant Dube, Rashmi Sharma
Certificate : **Link**

Volunteering Work

Edumonics Pvt. Ltd., New Delhi, India *Jun 2021 – Oct 2021*
Content Researcher

Research content and corrections, Blogposts, designed landing pages for the website, digital marketing along with SEO techniques to boost up the reach.
Certificate: **Link**

Muskurahat Foundation, NGO, India *May 2023 – Aug 2023*
Fundraising Volunteer
Raised funds of 8000 Indian Rupees, learnt how NGO's function, their causes and the situation of children in our country, how to approach people for fundraising, awareness drives
Certificate: **Link**

Positions Of Responsibility

Ambassador of Design -HEIV (The scientific community)
Design Head - The Indian Game Theory Society
Design Head - Society of Robotics (DTU)
Design Head - DEPTH (Engineering Physics Society)

- Executed visual designs for print and digital media including brochures, advertisements, websites and social media graphics.
- Responsible for leading a team of designers and for overseeing all aspects of the design process from start to finish.
- Created both original designs and utilised existing templates to produce high quality work that met the team's branding guidelines.

Awards & Achievements

Received the **India Merit Scholarship** worth £5,000 for my studies at The University of Edinburgh
School Topper in Class X Board Examinations
Academic Head Boy of the school in class XII
CS50X, Harvard University **Certificate**